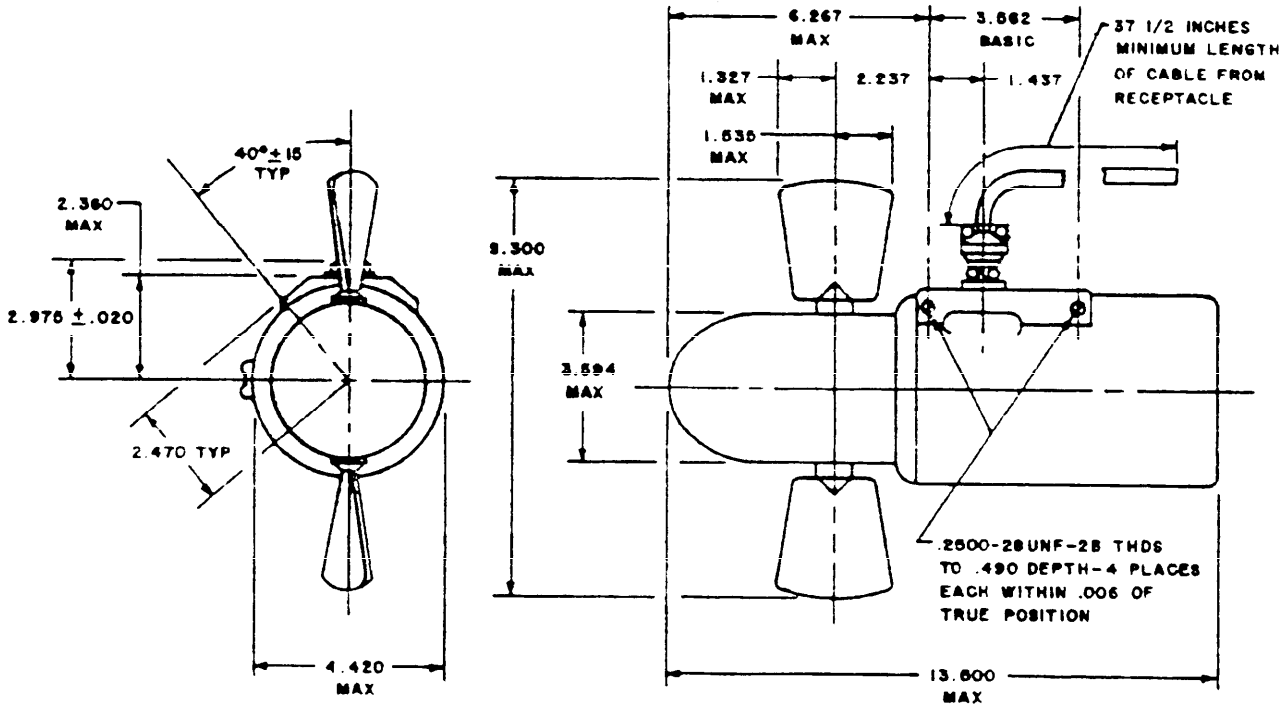
MIL-G-21480/9B(AS)
6 March 1981
SUPERSEDING
MIL-G-21480/9A(AS)
22 September 1971

MILITARY SPECIFICATION SHEET

# GENERATOR SYSTEM, EMERGENCY, 1.7 KVA, 400 HERTZ, ALTERNATING CURRENT, RAM AIR TURBINE DRIVEN

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the generator described herein shall consist of this document and the latest issue of specification MIL-G-21480.




| GENERATOR DATA | |
|---|---|
| RATED VOLTAGE AC | 120/208V |
| RATED OUTPUT AC | 1.7 KVA |
| PHASE | 3 |
| FREQUENCY | 400 HERTZ |
| SPEED, RPM | 12,000 |
| RATED POWER FACTOR | 0.9 LAG |
| EFFICIENCY, MIN AT RATED LOAD | |
| MAX WT | 25 LBS |
| CLASS | C |

| MILITARY PART NUMBER |
|---|
| M21480/9-1 |

Figure 1. Generator

FSC 6115

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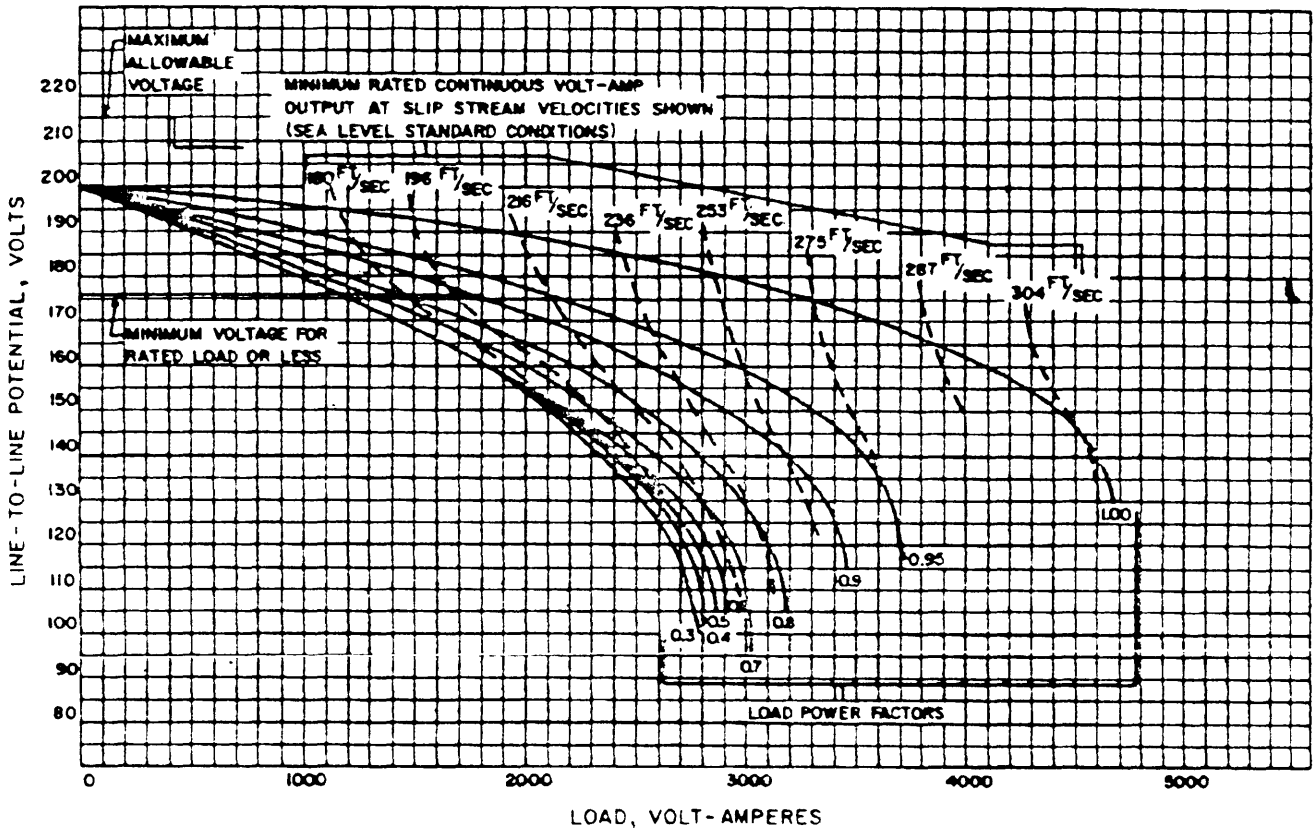



Figure 2. Potential Vs. Load



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REQUIREMENTS

1. General. The generator shall be in accordance with MIL-G-21480 and as specified herein. (Where conflict exists between MIL-G-21480 and this specification sheet, this specification sheet shall apply).

2. Configuration. The configuration shall be as shown in Figure 1.

3. Mounting provisions. Mounting provisions shall be as shown in Figure 1.

4. Operating conditions. The generator shall operate satisfactorily when subjected to load factors encountered in airplane maneuvers as follows:

| | |
|---|---|
| Vertical down: | 7g |
| Vertical up: | 4g |
| Side: | ±2g |
| Fore & Aft: | 6.5g |

5. Consolidation of control components. This requirement of MIL-G-21480 is not applicable.

6. Manual system control. This requirement of MIL-G-21480 is not applicable.

7. Lubrication. The generator bearings shall be of the sealed, factory-lubricated type and shall conform to MIL-B-7949. The generator shall not require lubrication prior to installation, nor during its service life which will include approximately 500 cycles of pressure change from sea level to 50,000 feet.

8. Flexible drive and shear section. This requirement of MIL-G-21480 is not applicable.

9. Electrical connections. The generator shall be provided with an output cable as shown in Figure 1. A suitable strain relief clamp shall be provided. The phase sequence shall A-B-C.

10. Color coding. The leads of the output cable shall be color-coded as follows:

| | |
|---|---|
| Phase A | Red |
| Phase B | Blue |
| Phase C | Yellow |
| Neutral | Black |



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11. Cooling. The generator shall be air-cooled. All necessary cooling shall be accomplished by the normal flow of air over and through the generator.

12. Life of generator. The generator shall have a minimum life of 1,000 flight-hours to first overhaul during which it shall be capable of operating for 250 hours for intervals of 2 hours minimum.

13. Magnesium alloy parts. Magnesium alloy parts shall not be used.

14. Voltage limits. The generated phase voltage, nominally 115 volts rms, shall not exceed 124 volts rms, nor be less than 102 volts rms, over the airspeed and load range as specified herein for all steady-state conditions. Rated 0.85 PF lagging load applied and removed shall not cause the voltage to be greater than 124 volts rms, nor less than 102 volts rms, line-to-neutral, for longer than 0.2 second over the range of airspeeds specified.

15. Voltage modulation. The voltage modulation requirements of MIL-G-21480 are not applicable.

16. Phase displacement and voltage unbalance. A 1/4 load unbalance shall not cause more than 6 percent voltage unbalance. A 1/5 load unbalance shall not cause more than a 4 percent voltage unbalance.

17. Waveform. The phase voltage waveform shall be within the following limits for all normal operating conditions of the system:

    (a) Crest factor: 1.41 $\pm$10%

    (b) No single harmonic is to exceed 5% of the fundamental.

18. Load capacity. The system shall be capable of delivering rated load continuously. Also, the system shall deliver 150 percent rated current for 5 minutes at airspeeds into the turbine of from 180 KIAS to Mach 1 and 200 percent rated current for 5 seconds at airspeeds into the turbine of 180 KIAS to Mach 1 without damage.

19. Starting and warmup. With full load connected, the generator shall perform within specified limits for operation within seven seconds after the system is dropped into a 110-knot airstream and in five seconds after the generator is placed in a 180-knot airstream. No warmup shall be required for the proper operation of this system.



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20. Nameplate data. The nameplate shall conform to MIL-STD-130 and contain the following information on the nameplate:

Generator System, Emergency, 1.7 KVT, 400 Hertz,
Alternating Current, Ram Air Turbine Driven
Federal Stock Number
M21480/9-1
Manufacturer's Name (no address)
Manufacturer's Part Number
Contract Number
Government Acceptance Stamp
Acceptance Date
Serial Number

21. Terminal markings. The terminal marking requirements of MIL-G-21480 are not applicable.

22. Sampling, inspection and tests. Sampling, inspection, and tests to determine conformance to this specification shall be conducted as specified in MIL-G-21480 and as specified herein.

23. Control components. The test requirements of MIL-G-21480 for control components are not applicable.

24. Flight test. 100 hours flight time with the generator in operation shall be required for qualification of the generator.

25. Mounting. During testing, the generator shall be mounted in the normally installed position in the aircraft.

26. Airflow. The generator shall be subjected to a free airstream of 110 KIAS at a temperature of 25 $\pm$ 15°C. Cooling of the generator shall be accomplished only by ram air.

27. Maximum speed for regulation. The generator shall be operated at air velocities from 133 to 200 MPH and with loads varied over the load range of the generator. The voltage shall remain within the limits specified in Requirement 14.

28. Overload. The ability of the generator to deliver the following loads shall be demonstrated at sea level, 50,000 feet and 65,000 feet. During qualification testing the generator shall also be operated at test condition A and at sea level with the inlet air at -55°C. Between demonstrations the generator shall be run at rated load for at least one hour. The voltage shall remain within the limits specified in Requirement 14 at no load and rated load as well as at the following overload conditions:



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(a) 150 percent rated load current for five minutes.

(b) 200 percent rated load current for five seconds.

The generator shall be capable of furnishing overloads as shown in Figure 2.

29. Unbalanced loads. With the generator carrying a balanced three-phase load of 430 VA, the generator shall be capable of handling an additional load of 125 VA added to any phase.

30. Efficiency. The efficiency of the generator shall be measured at rated voltage and rated load at minimum, average and maximum generator speeds.

31. Endurance. The service life of the generator shall be at least 250 hours and shall be demonstrated by a suitable endurance test.

32. Operation at -65°C. After 25 hours at -65°C, the generator must start and operate within 30 seconds after minimum airflow is reached at rated load.

33. Flight test. A 100-hour flight test shall be conducted on an A-4 aircraft. A minimum of 10 actuations and a minimum of 20 operating hours during this period is required over the IAS and altitude operating range. At least two hours of the test shall be during precipitation.

34. Switching operation. The switching operation test of MIL-G-21480 is not applicable.

35. Protective operation. The protective operation test of MIL-G-21480 is not applicable.

36. Standard performance. The standard performance test of MIL-G-21480 is not applicable.

37. Salt spray. All protective covers except hermetically sealed covers shall be removed and the generator shall be salt spray-tested for 100 hours according to Procedure I of MIL-E-5272.

38. Shock. The generator shall be subjected to the Shock Test of MIL-E-5400. Following this test, the generator shall deliver rated load for two hours and shall maintain rated voltage.

39. Flexible drive. The flexible drive test of MIL-G-21480 is not applicable.

40. Drive shear section. The drive shear section test of MIL-G-21480 is not applicable.



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NOTES:

1. Dimensions in inches. Unless otherwise specified, Tolerances: Decimals ± .031, Angles ± 1°.

2. Where only maximum dimensions are shown the part need not have the shape shown, but shall be contained within the outline shown.

3. The word generator will refer to the entire assembly of ram air turbine and generator.

Preparing activity:

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(Project No. 6115-N464)

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DOCUMENT IDENTIFIER AND TITLE

MIL-G-21480/9B(AS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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